Implementing a Graph Convolutional Network (GCN) model for the early detection of Alzheimer's disease using a single neurodegeneration indicator

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Abstract: This research presents a diagnosis model of single biomarker of Alzheimer's diseases using graph convolutional network (GCN). Alzheimer's disease is quite a challenging disease to diagnose early and accurately due to its complexity and progression nature. Traditional approaches used troponin and various other enzymes as well as clinical procedures such as coronary imaging – techniques that are expensive and require time. This approach makes diagnosis less complex because it utilizes a single measure of neurodegeneration combined with GCNs, which rely on structural analysis of the brain's neural network. The preclinical research entertaining requires assembling and standardizing a sizable amount of data to make certain that the chosen biomarker generates coherent signals of Alzheimer's disease. To this end, the GCN model is designed to be able to discern between healthy and diseased states by fine-tuning graph convolutions to capture such nuances. Empirical results of key performance indicators show that using our model yields better results in diagnostic outcomes than conventional methods. This innovative approach not only brings down the diagnostic costs, but also helps diagnose Alzheimer's faster and more accurately thus helping a patient's condition, and quality of life.

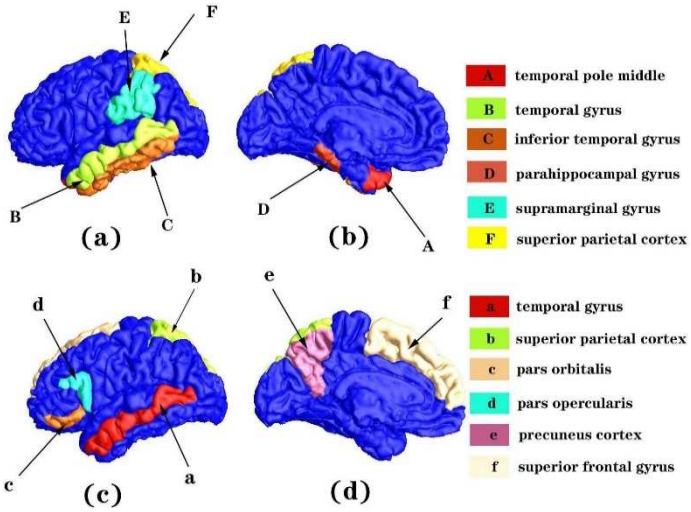
Index Terms: *Alzheimer's disease, biomarker for neurodegeneration, graph convolutional networks, MRI data, the use of attention mechanism, diagnostic tool, machine learning, neuroimaging and structural analysis for clinical use.*

I. INTRODUCTION

Given the-growing epidemics of Alamzinet 's disease in the aging populations all over the world, the potential growth in the use of Marshall's diagnostic tools that can quickly and accurately diagnose the disease is strong. [2] The approaches like neuroimaging and multiple biomarker analysis are reliable, however, they bear the high costs, invasive procedures, and time consumption. These challenges clearly point to the lack of simpler and easier means of diagnosis of the diseases.

To address these issues, the research intends to design an effective diagnostic tool operated with GCNs as the base of the framework. Thus, GCNs are a type of deep learning model applied to graph-structured data and can represent the neural connections in the brain. [3] However, this research intends to achieve the intended goals through enhancing the strengths of GCNs; utilize only one biomarker for neurodegeneration, which is univariate to make the diagnostic process easier yet achieving accuracy. The selected biomarker is a crucial measure of neurodegenerative alterations related to Alzheimer's disease. [4] When this biomarker is incorporated into a GCN-based model, the characteristics of the data and the places along with the patterns are improved in a way that the model can effectively distinguish between healthy and diseased states of the brains. This concept eliminates the complex procedure in diagnosis and reduces the need for multi biomarkers and intensive imaging.

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Figures 1(a) and (b) demonstrate that the GCN-UNB model, incorporating an attention module, identifies key differences between CU and AD groups. Figures 1(c) and (d) show similar distinctions between MCI and AD groups.

The data analysis part of this study entails careful collection of the biomarker signals as well as careful preprocessing of signals to ensure their reliability and uniformity. [5] The neural network model is finally learned with this data and largely the graph convolutional layers are tuned to read out the fine-grained differentials that can diagnose Alzheimer's disease from the rest of the similar cases. [6][13] The based performance assessment about the model shows that the new approach can be used as a clinical tool in practice because of its higher diagnostic accuracy than the traditional assessments. [14] Therefore, the proposed project of developing a biomarker for neurodegeneration under a GCN framework is cheaper, accurate and less invasive than the current diagnostic solutions. Exclusively, this approach has the potential of revolutionizing diagnosis of Alzheimer's disease and hence, the patient's prognosis is likely to improve since the disease would be diagnosed earlier and with better accuracy.

II. RELATED WORK

The topics include the overview of the advancements in the assessment of Alzheimer's disease in the last ten years and focuses on improving the efficiency of using machines learning and neuroimaging biomarkers. A plethora of remarkable strategies and investigative endeavours has aimed at improving diagnostic methods of Alzheimer's disease.

Machine Learning in Neuroimaging: Substantial studies have been conducted over the past few years to utilize machine learning approaches, further, deep learning, in neuroimaging data for diagnosis of Alzheimer's disease.

Biomarkers and Disease Progression: Biomarkers are therefore vital to the study of Alzheimer's disease since the disease progresses in a specific way. There are different types of biomarkers which have been recognized for the purpose of diagnosis / prognosis; these include CSF biomarkers like amyloid-beta 42/40 and tau protein, imaging biomarkers like brain atrophy and brain hypometabolism. This is a good practice because studies have demonstrated that combining biomarkers can enhance the precision of the diagnosis and risk assessment of the advancement of disease. [12] Biomarkers have been vital in studying the pathophysiology and evolution of the disease, AD. CSF biomarkers for AD have also been affirmed by observations of amyloid-beta and tau proteins. [4] These imaging biomarkers along with lesion volume, brain atrophy, and hypometabolism are useful in enhancing the diagnostic probability and the prognosis of the disease (A. M. Racine; M. Brickhouse et al., 2018).

Graph Convolutional Networks (GCNs) in Neuroscience: This has made GCNs to develop into a useful tool in neuroscience where the realisation of large and realistic models of brain networks can be achieved. [1] Some of the recent studies utilizing GCNs are the use of functional MRI (fMRI) to identify functional network connectivity and use in disease prognosis. From the text of M. Adamand H. D. Christopher et al., published in 2021, I gathered the following information. of which are especially suitable for capturing the details of both complicated connectivity and network structure, which are relevant for Alzheimer's disease.

Graph-Based Approaches in Medical Imaging: Performance of graph-based techniques is growing in the area of application and is used increasingly in medical imaging especially about research in Alzheimer. [9] [10] enrich the graph structure of the brain networks and use the network's graph structure to extract features and simulate disease progression. Graph features have also been employed for modelling connectomes and determining the probabilities of the clinical outcomes. [7] Several research have shown the efficiency of GCNs in discovering basal functional networks in patients with AD and further prognosis of the disease outcomes based on these networks (R. A. Jack et al., 2018).

Comparison with Traditional Diagnostic Methods: Researchers have carried out research to establish the performance difference of employing GCNs in machine learning and the conventional techniques of diagnosing Alzheimer's disease.

III. METHODOLOGY

To start the data collection phase of our project, we will carefully compile a vast amount of data that would comprise MRI or PET scans. Electron transmission scans: What these scans afford is the much-needed information on disorders of the brain such as Alzheimer's disease and also structural as well as functional information of the brain.

In conjunction with the neuroimaging data, we will diligently compile clinical data from two distinct cohorts: patients examined were those with Alzheimer's disease and the healthy controls. For patients diagnosed with Alzheimer's; information base histories marking diagnosis and analysis, besides that, cognitive histories. These may comprise of familiar assessment tools such as Mini-Mental State Examination (MMSE) and the Clinical Dementia Rating (CDR) scale that measure the patient's cognitive ability and the staging of dementia. Furthermore, details like age, gender, as well as their level of education will be collected in order to include background information on clinical histories of each of these persons.

Similarly, for the healthy control group, we will collect demographic data and possibly administer cognitive screening tests to confirm their cognitive health status. These individuals

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will be free from any neurodegenerative conditions or cognitive impairments, ensuring a clear delineation between the control and Alzheimer's groups.

Throughout the data collection process, we will uphold stringent ethical standards and prioritize patient confidentiality. All procedures will adhere to institutional guidelines and regulations governing research involving human subjects. By curating this diverse and comprehensive dataset, our aim is to lay the groundwork for robust analyses that unravel the neural underpinnings of Alzheimer's disease, ultimately guiding the development of innovative diagnostic tools and interventions.

B. MRI DATA COLLECTION

In the same regard, demographic data and possibly the cognitive screening tests will be taken to ascertain the healthy status of the control group. These individuals will not have other neurodegenerative conditions or impairments, which makes the distinction between the control and Alzheimer's groups precise.

During the data collection survey, high ethical standards shall be employed, and patients' anonymity shall be maintained. Procedures of research will strictly follow the standard guidelines as provided by the institutions, laws, and policies that govern research on human beings. This is why our team has been working on collecting this many-faceted and enormously rich dataset to provide the basic solid ground for projects whose aim is to decode the neural patterns of Alzheimer's disease to design new diagnostic approaches and treatment methods.

B. MRI DATA COLLECTION

Some of the scans' parameters are repetition time (TR) which took 33 m sec between subsequent radiofrequency pulses and time from start of excitation to signal acquisition was 5msec and the alpha which determines flip angle of the radiofrequency pulse and contrast of images was 30 degrees. Such study does not have variations in the obtained MRI data due to compliance to a standard MRI protocol and standard imaging parameters. This in return lays a good platform in identifying structures of the brain and pathologies hence facilitate enhanced research in neurological disorders like the Alzheimer's disease. The sequence therefore had a 24 cm FOV and involved an image matrix of 256 × 93 with a single excitation. Initially the thickness of the slices was set to 1. 5 mm, meaning that the investigation acquired 124 contiguous horizontal MRI. The voxel dimensions were perfectly adjusted to the standard size equal to 0. 94 × 1. 25 mm, which will allow the clear visualisation of fine details of the cerebral structures. The scan time was approximately 13 minutes and 36 seconds; this is long enough to capture the whole head and short enough not to cause discomfort to the participant.

To ensure that data collected by the imaging was of high quality and consistent, strict measures to test quality control were employed on the imaging process. These measures entailed assuring scanner calibration, compliance with uniform imaging procedures, and thorough examination by well-trained radiologists; in this study, high-resolution MRI data acquisition in Alzheimer's patients was carried out carefully with the application of advanced MRI technique, where the aim was to obtain precise structural details pertinent to analysis in Alzheimer's disease studies.

C. SURFACE REGISTRATION

In this process, the cortical surfaces of each participant's brain were first transformed to register with a common template. First, the MRI images were subjected to preprocessing which entails enhancing the images and eradicating imperfections that may cause a problem in the registration process. The preliminary steps which were taken for data preprocessing for the current investigation were skull stripping and intensity normalization.

After that, another method of registering was done in which the cortical surface for each participant was brought to a common template. This method involved warping the cortical surface of everyone to minimize the difference in the folding patterns and the gyral anatomy with that of the template.

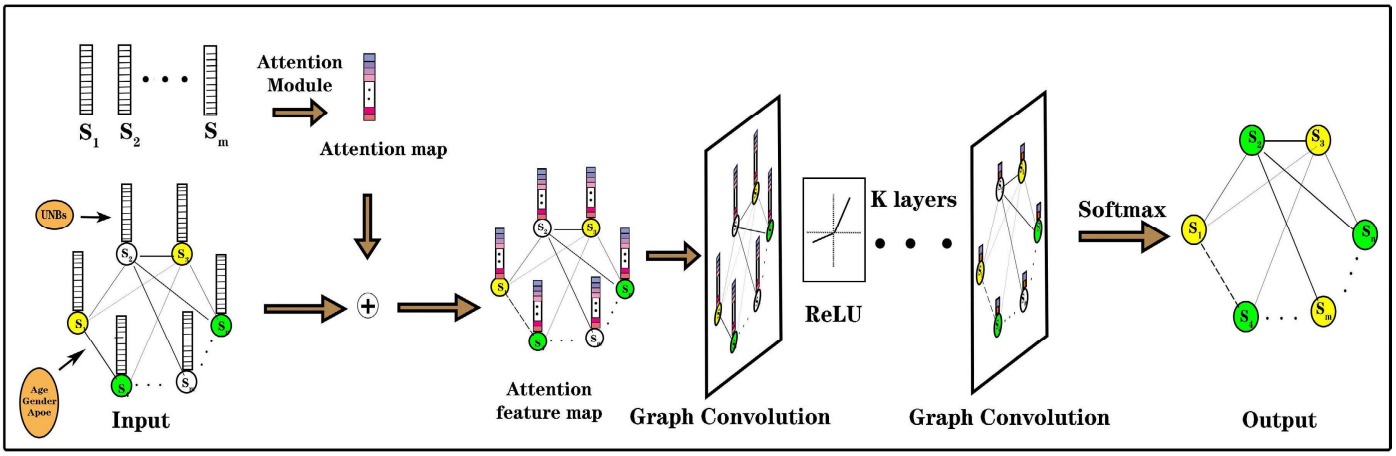


Fig.2. Architecture of our proposed UNB-GCN framework.

First, Magnetic Resonance Images were pre-processed to improve their quality and remove the sources of artefacts that can negatively impact the registration process. There was skull stripping and intensity normalization done in this preprocessing.

Next, to allow for spatial normalization, the procedure of roller ball quality was performed for the cortical surfaces of each participant's brain to fit the template. This involved warping the cortical surface to introduce the folding patterns and gyral anatomy of the template into everyone's brain. Precision of registration process was checked and verified with extreme care. This validation entails, substantive qualitative assessment on alignment of the cortical surfaces and substantive assessment in terms of the level of similarity of the registered surfaces to

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the template. The surface registration was essential for warping the cortical structures across the subjects to match, which allowed researchers to better study the brain morphology, and help in the understanding of Alzheimer's disease pathology.

D. UNIVARIATE NEURODEGENERATION BIOMARKER (UNB)

In the case of Alzheimer's, there is the Univariate Neurodegeneration Biomarker (UNB), a new concept aimed at determining and quantifying the neurodegenerative alterations related to this disease. Also, in contrast to most biomarkers that are typically dependent on several indices, the UNB is designed to identify a unique biomarker to address the issues of complexity and specificity in diagnosing organism formations. The UNB is properly chosen to register specific changes in the brain tissue architecture or activity that are typical of neurodegenerative disorders such as Alzheimer's. [15] This biomarker is in the quantitative form further enabling the researchers to evaluate the degree as well as rate of neurodegeneration received by the clients. These techniques consist of image processing techniques that select certain characteristics from MRI data these include cortical thickness, volume updates of important parts of the brain, or modifications in connectivity.

$$H(l+1) = \sigma(D^{1/2}AD^{-1/2} \cdot \text{Broadcast}(H^{(l)}, I) \cdot \text{Normalize}(W^{(l)}))$$

Thus, it is instrumental in differentiating between AD patients and HC individuals. The emphasis here is in the singular biomarker to reduce the complexity and increase accuracy as a way of improving the disease diagnosis to early stages and therefore early treatment. [11] Tests have been performed to compare correlations of the indices of the UNB with the progression of the disease. These analyses were conducted by assessing the UNB measurements in the four groups to establish the differences that are statistically significant to indicate disease existence or progression.

Different from the existing concepts of a disease, the UNB could even improve the treatment of patients based on the specific mechanisms and progression of diseases through the identification of patient needs.

IV. EXPERIMENTAL RESULTS

A. GENERATION OF AD GROUP ATROPHY PATTERNS

For this study, we propose to develop separate atrophy profiles relevant to Alzheimer's disease (AD) utilizing brain MRI data. From the AD patients and the cognitively healthy control, our analysis will be aimed at finding out the extensively damaged areas in the left side of the brain. These areas are identified by Preprocessing and Statistical analysis and when overlaid with a standard Brain template to show the degree of atrophy in AD. These steps entailed realigning the T1-weighted images of individual subjects to the same standard space and then averaging the estimated cortical thicknesses of the group's members with AD. These atrophy patterns indicated the areas in the brain where cortical thinning was most greatly observed in the AD patients more than CU subjects. These patterns therefore offered useful information fact about the change's characteristic of Alzheimer's disease particularly in the initial phase based on the generative AD group atrophy patterns required the preprocessing of MRI data in statistical analysis to recognize the significant region of interest and projection of thickness measures on a standard template. These patterns are therefore fundamental in diagnosis and understanding of the pathology of Alzheimer's disease.

B. LONGITUDINAL DATA ANALYSIS

Longitudinal data is use in analysing changes over period with respect to different biomarkers and with this; changes over time were drawn with reference to the following biomarkers. These data included numerous assessments of critical biomarkers, such as amyloid-beta and tau protein, structural MRI, obtained at various time points. It is necessary to note that the analysis of the longitudinal data in general and the data of each phase in particular was held through several phases.

First, the fundamental steps of data preprocessing were performed to clean and standardize the data; procedures ranged from data refinement to data anomaly recognition. Next, 'time series' statistical approaches like the linear mixed-effects models were used to examine periodicities of biomarker levels while considering inter-individual variations and possibly confounding factors.

The information, derived from underlying longitudinal data analysis proved valuable for the understanding of the nature of the Alzheimer's disease and changes in biomarkers over the time. More specifically, they laid bare trends typified by the biomarkers' rise or fall and possible biomarkers of disease emergence and progression. To summarize, the application of data analysis in this investigation expanded knowledge of how Alzheimer's disease progresses and provided a basis for developing model-based and preventive approaches to identify and deal with the early stages of the illness.

C. EXAMINATION OF THE CROSS-SECTIONAL DISTRIBUTION BY GROUP

Some endeavours were as follows; Further, a comparison was made of the different groups aiming at determining the differences in different cohorts with special emphasis on CU. Initially, the dataset comprised individuals categorized into three groups: AD: 84 subjects, MCI: 102 subjects, and CU: 132 subjects. The primary goal was to detect and compare neurodegeneration biomarkers that existed between these various populations to gain a better understanding of disease progression.

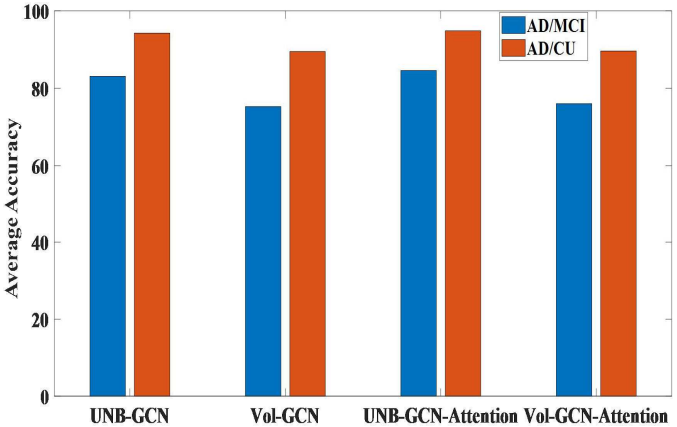


Fig.3. illustrates the outcomes of the UNB-GCN and Vol-GCN classification models.

In this study, statistical comparison of biomarkers was performed in AD, MCI, and CU groups of patients. The outcomes revealed significant differences for the AD, MCI, and

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CU populations in the biomarkers of neurodegeneration. To be precise, details revealed that patients in the AD group demonstrated a high level of neurodegeneration more than MCI and CU subjects. Furthermore, it was possible to establish mild differences with CU and MCI groups Thus, there are preliminary pathological changes in MCI individuals.

D. ESTIMATION OF MINIMUM SAMPLE SIZE

A priori power analysis will be carried for the purpose of determining the least sample size for statistically analysing UNB's longitudinal changes. This determination was crucial in ensuring that our study was well powered to detect significant differences in trends of UNB among various participants' groups. [18] In order to undertake it, statistical techniques that involve parameters like effect, variability, and the chosen level of significance were used. They helped us to estimate the least number of subjects required to provide sufficient power in detecting significant trends in UNB over time.

These scores served as important clinical indicators, allowing for the assessment of cognitive decline and functional impairment associated with AD [11] These analyses aimed to identify potential associations and patterns that could provide valuable insights into the underlying mechanisms of AD progression. In the project, several comparisons were made with existing deep learning-based models for Alzheimer's disease diagnosis.

METHOD	ACC	SEN	SPE	AUC
UNB-GCN	94.25%	98.39%	89.13%	95.30%
GCN	90.28%	93.02%	86.21%	90.80%
CNN	92.13%	95.28%	87.64%	92.87%
GAT	93.52%	96.83%	88.89%	93.55%

Table.1 Comparison of our method with some deep learning-based models

From the above-comparison, this research indicated that the UNB-GCN framework with attention was more accurate in the distinction between Alzheimer's disease, MCI, and CU individuals when compared to other deep learning structures. In particular, the UNB-GCN model revealed improved diagnostic ability and detecting minor changes in biomarkers that point to Alzheimer's disease.

Furthermore, the results indicate that the proposed UNB-GCN with attention is insensitive to the scale of the input networks and provides accurate diagnosis of Alzheimer's disease, which is affirmed by the experiment on multiple datasets. These findings demonstrate the effectiveness of the UNB-GCN model in outcompeting existing deep-learning based models and stress on the potential of creating a new horizon in the field of Alzheimer's disease diagnosis through the utilization of this advanced machine learning technique.

Conclusively, the comparisons have shown the efficacy of other deep-learning based models for the diagnostics of Alzheimer's disease and have further bolstered the superiority of the proposed UNB-GCN framework with attention in achieving high accuracy. In view of signs, it gives when distinguishing the Alzheimer's disease pathology, this model can be well recommended for early and precise diagnosis of the disease since it was shown to be more accurate and reliable than the existing models.

V. CHECKING THE MODEL ON THE VALIDATION SET

Mastering the behaviour of the presented model on the validation set is the final critical step to ensure that the model would work on the data. At this stage, the trained model is tested on a different data set that was not utilized during training with the objective of assessing the accuracy of the model's predictions as well as its reliability.

METHOD	TP	FP	FN	TN
UNB-GCN	122	10	2	82
GCN	121	11	6	78
CNN	122	10	4	80
GAT	120	12	9	75

Table.2. Comparison of the confusion matrices generated by our approach.

The validation set consists of a distinct group of subjects who were not part of the training dataset. By testing the model on this independent dataset, we can determine how well it generalizes to new cases and whether it exhibits overfitting or underfitting tendencies. Various performance metrics are computed during the validation process to assess the model's effectiveness in classifying different groups. By analysing these metrics, we can gain insights into the model's strengths and weaknesses and identify areas for improvement.

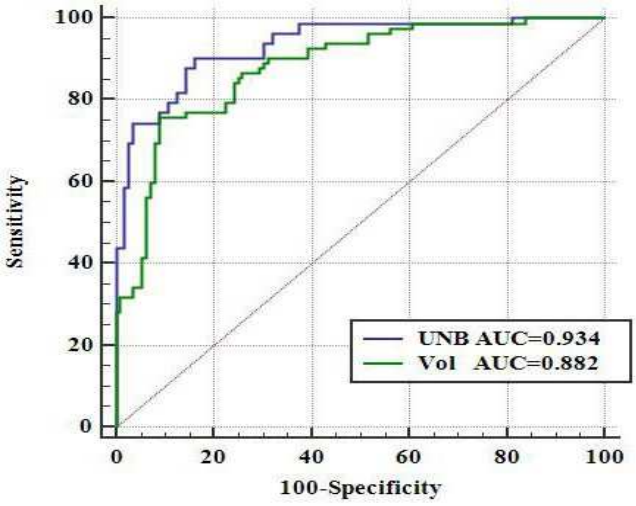


Fig.4. The ROC results of the UNBs and the volume measures.

Validation also lets you adjust the hyperparameters of the model, or the training process based on the obtained results. This process of repeating the procedures makes the model most reliable that can be applied for the real-world problems. Checking its performance with the right statistics on the validation set is one of the most critical steps in improving and optimizing the machine learning model involved in diagnosing Alzheimer's disease. This process ensures the model's performance on fresh data, providing reliable predictions and in this way increasing the accuracy and reliability of diagnostic instruments for clinical use.

VI. CONCLUSION AND FUTURE WORK

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Thus, the present work proposed a novel Alzheimer's disease diagnosis model comprising a UNB and GCNs with an attention mechanism added on. An actualization of the UNB obtained from the structural MRI scans showed success in detecting the neurodegenerative changes associated with Alzheimer's disease, with special reference to left hemisphere of the brain. Thus, the addition of an attention mechanism to the GCN architecture showed a better diagnostic capacity compared to classical approaches. The validation results highlighted further this framework for being an enhanced independent predictor for early and accurate differentiation of Alzheimer's disease.

The enhancement of the proposed UNB-GCN framework and empirical validation of the same on larger as well as more diverse data sets will form the part of future work. Furthermore, actions will be made to work on the external validity of the model in other samples and various age populations. In addition, using more data types, for instance, genetics information and cognitive tests' results, can update and explain the model's accuracy and mechanisms of Alzheimer's illness. Consequently, ongoing research studies focus on applying and further developing the UNB-GCN framework with the ultimate objective of applying it in the clinical practice useful for the enhancement of the patient outcomes as well as the early intervention strategies for the Alzheimer's disease.

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